Sample data (synthetic), Jul-Sep 2026.

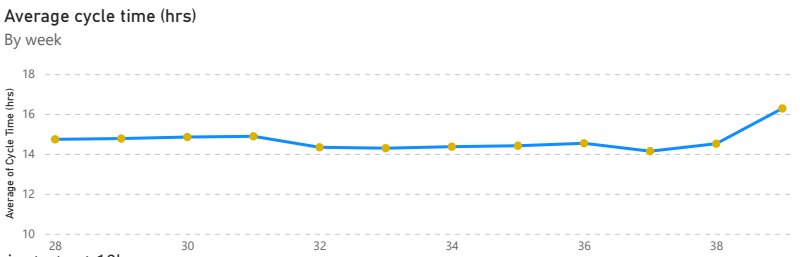
Overview

Peak Period:
Week 39 (21- 27 Sept)
Order volume ~50% above normal.

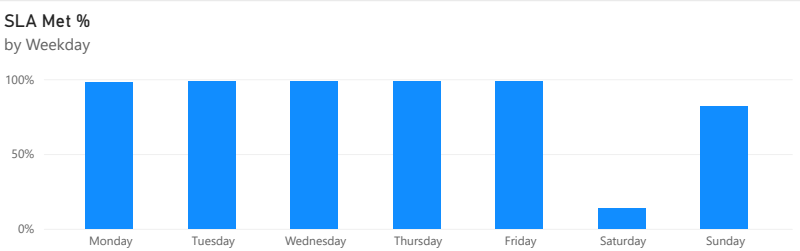
Order Cycle Time

SLA Met %
86.9%

Average of Cycle Time (hrs)
14.68



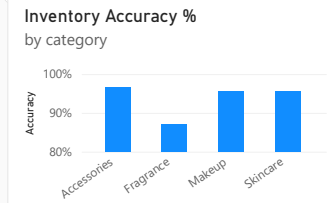
Axis starts at 10hrs



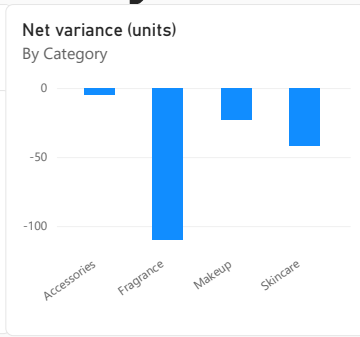
Key Finding:
Only **14% of Saturday** orders meet **SLA,** and Saturday accounts for **77% of all late orders**

Inventory Accuracy

Inventory Accuracy %
93.7%



Axis starts at 80%



Fragrance Net Shortage (Units)
Sum of Variance
-110

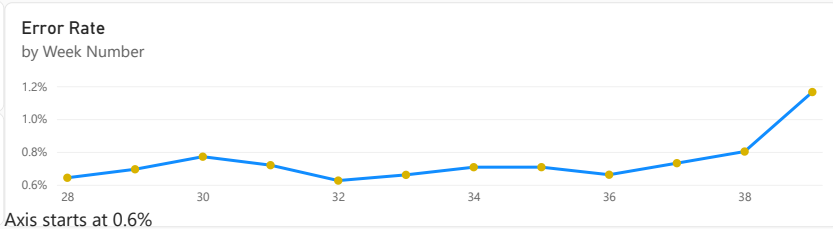
Fragrance Net Shortage (SGD)
Variance Value (SGD)
-15.22K

Key Finding:
Fragrance Inventory accuracy is **~9 points below** other categories *(87% vs ~96%)* and carries the **largest net shortage, about S$15.2K**, accuracy fell furthest in the peak week

Productivity

Error Rate
0.76%

Lines per hour
9.12

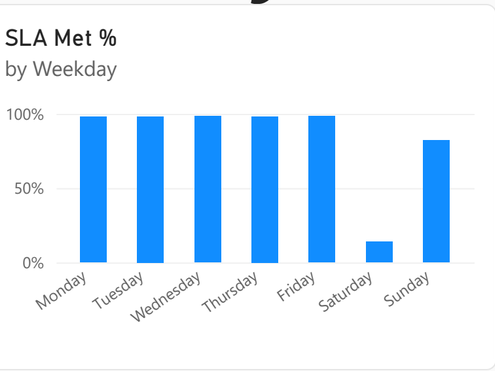


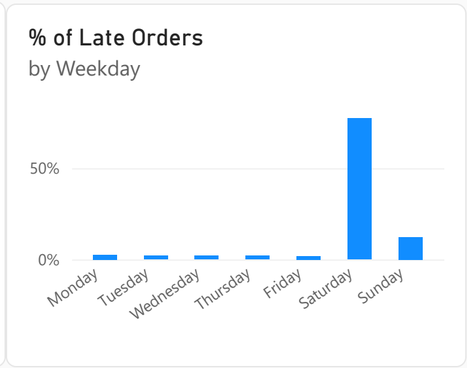
Axis starts at 0.6%

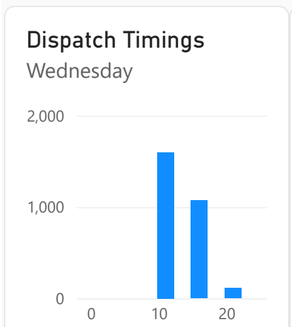
Key Finding:
Error rate rose from ~ **0.7% to 1.2%** in the peak week, alongside temp staff and overtime

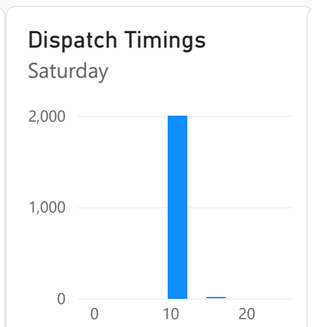
Order Cycle Time

Sample data (synthetic), Jul-Sep 2026.

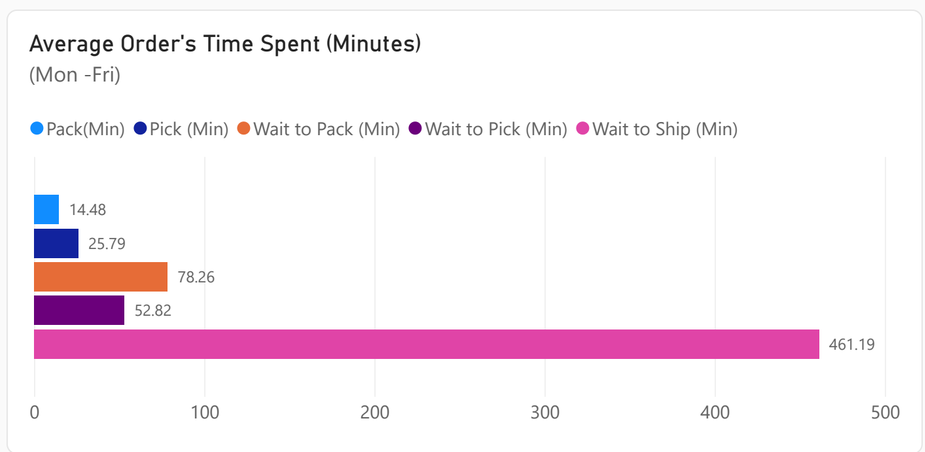












Problem :

- Only **14% of Saturday orders meet SLA** (vs ~98% on weekdays), averaging ~38 hours to ship
- Sunday orders also wait for Monday's dispatch, but the wait is shorter, so most still ship within the 30-hour target
- Saturday causes **77% of overall Late Orders**

Cause:

- **Weekdays** have up to **3 dispatches,** while **Saturday only has 1**. Hence, orders on Saturday *received after ~10:00* miss the single 11:00 dispatch and have to **wait until Monday.**

Where time goes:

- Only **~6%** of an order's time is spent being **picked or packed**.
- Longest controllable wait is for packing:
  - Pack: ~78 min of queue for ~14 min of work (**~ x5**)
  - Pick: ~53 min of queue for ~26 min of work (~ x2)

Possible Options:

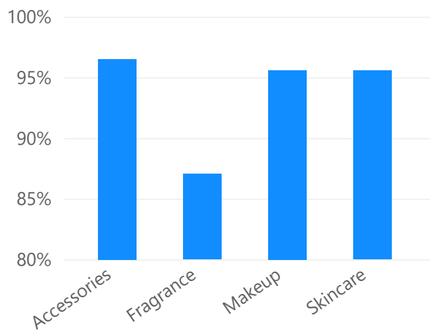
1. Add a ~21:00 Saturday dispatch.
   - Could **recover up to ~1,670 of ~1,800** late Saturday orders (~93%), at the *cost of Saturday evening labour.*
2. Investigate the packing queue (to confirm the cause first).
   - Review packer coverage on Mondays and 13:00-17:00, where the wait is longest.

Inventory Accuracy

Sample data (synthetic), Jul-Sep 2026.

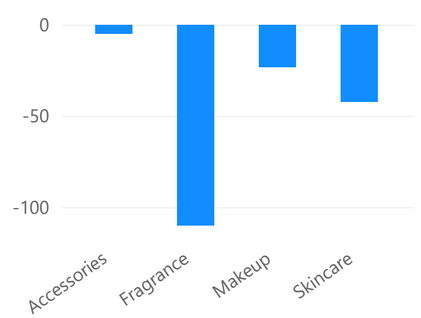
Inventory Accuracy %
93.7%

Inventory Accuracy %
by category



Axis starts at 80%

Net variance by category (units)



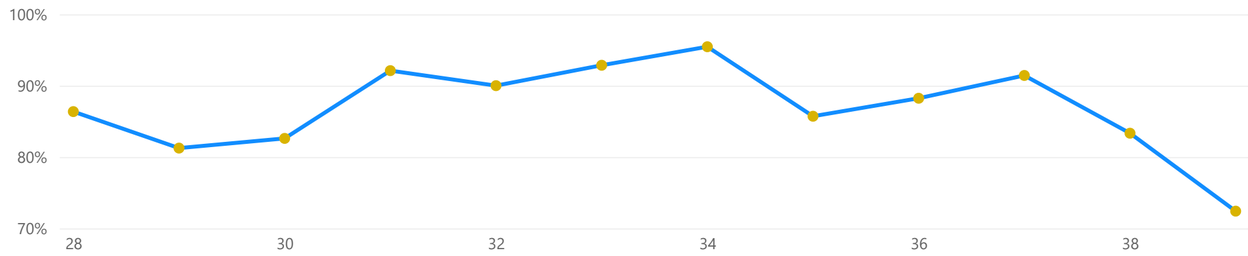
Fragrance Net Shortage (Units)

Sum of Variance
-110

Fragrance Net Shortage (SGD)

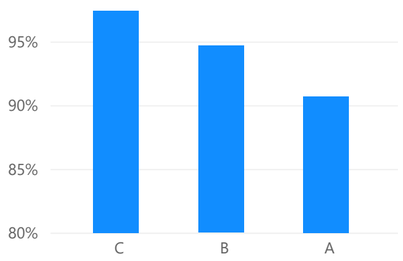
Variance Value (SGD)
-15.22K

Fragrance accuracy by week
By Week Number



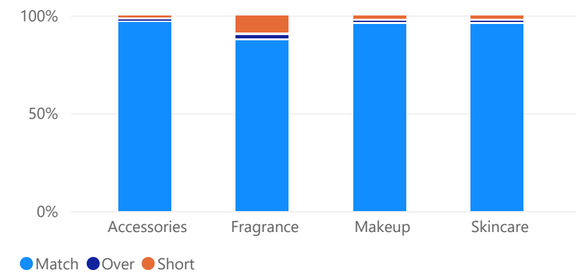
Axis starts at 70%, ~30 counts per week.

Inventory Accuracy %
by velocity_class



Axis starts at 80%

Short / Over / Match
By Category



Problem:

- **Fragrance** is the least accurate category at **~87%**, about **9 points below the others (~96%).**
- It also has the **largest net shortage: -110 units, about -S$15.2K** (in counted items).

Evidence:

- Most variances are **shortages** (the system shows more stock than is on the shelf), not random miscounts. **Fragrance** has the **largest "Short"** share (~10% of its counts).
- **Fragrance** accuracy fell to **72% in the peak week** (week 39, 29 counts), the lowest point of the period.
- **Fast-moving items** (class A) are also **less accurate**: **~91%** vs ~97% for slow movers.

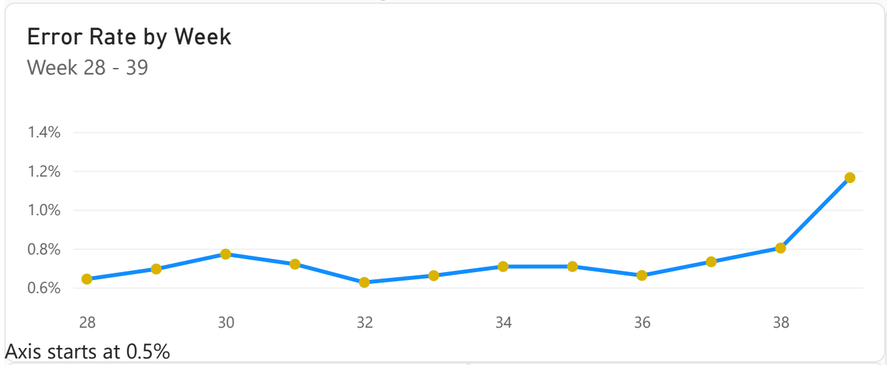
Possible options:

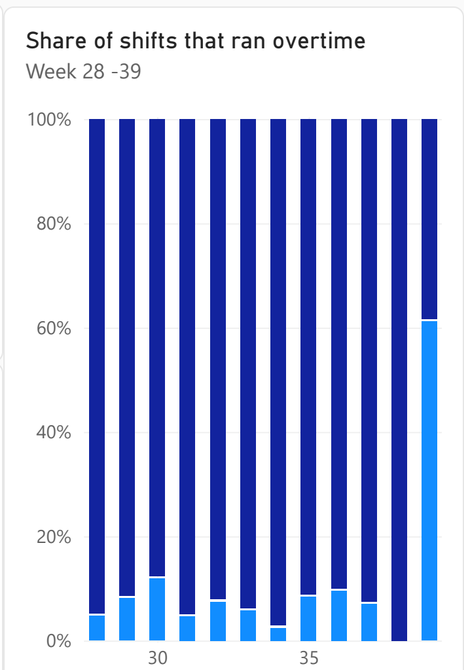
1. **Count Fragrance more often**, especially before and during peak weeks.
2. **Investigate the cause of the shortages** (to confirm first): review handling and access controls in the Fragrance zone, and the pick-confirmation steps.

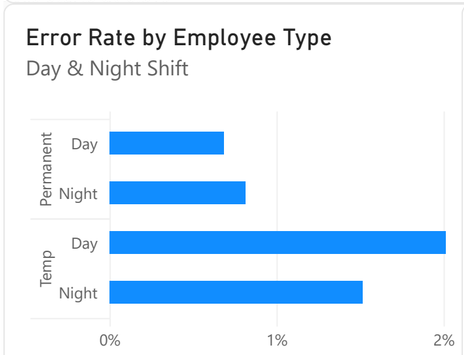
Productivity

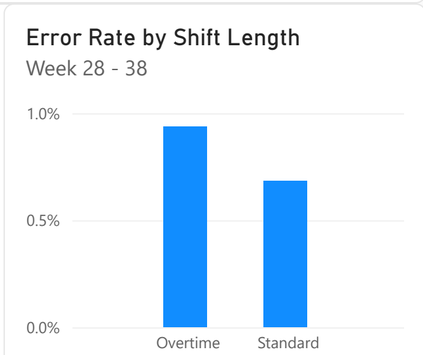
*Temps worked only in the peak week, so temp and peak-week effects overlap

Sample data (synthetic), Jul-Sep 2026.









Error Rate

Problem:

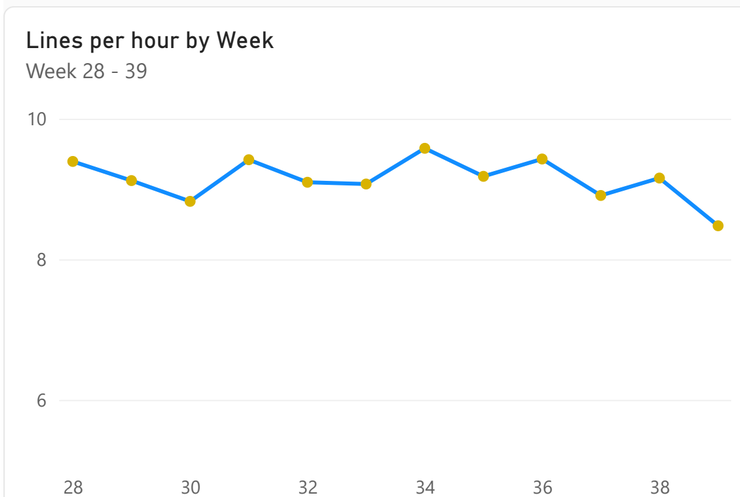
- Error rate rose from **~0.7% to ~1.2%** in the **peak week** (week 39), while **output barely moved** (**~9 lines per hour**, dipping to ~8.5).

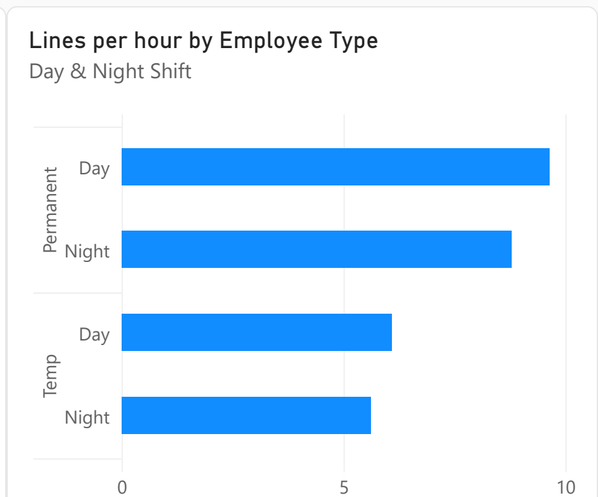
Evidence:

- **Temp staff** (joined 14 Sep) make **2-3× the errors** of permanent staff (~2.0% vs ~0.7-0.8%).
- **Overtime** carries more errors: **~0.95%** vs ~0.68% on standard shifts, even outside the peak (weeks 28-38). In week 39, **~61% of shifts ran overtime,** vs **~5-10% in normal weeks.**
- **Permanent staff also slipped** in week 39, so part of the rise is **peak-week pressure**, not only temps.

Possible options:

1. Train and onboard temps **before** the peak.
2. Cap overtime on quality-sensitive tasks, or add a QC check on overtime output.
3. Investigate why (to confirm first): the data shows a link, not the cause.





Lines Per Hour

Problem:

- Output held **steady at ~9 lines per hour in weeks 28-38**, and dipped only **slightly to ~8.5 in the peak week**, so speed was not the main casualty.

Evidence:

- **Temp staff** run at **~6 lines per hour**, about **35% slower than permanent staff** (~9-10).
- **Night shift** is slightly slower than day for both groups (~8%).

Possible options:

1. Plan for lower temp output when sizing peak-week capacity.
2. Review whether Night staffing or workload differs from Day (to confirm first).